



MANUAL

Limoxis Observer — Setup and user guide

From day one to everyday operation

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Introduction

Purpose

Limoxis Observer brings a hospital's day-to-day infection prevention and control work into one environment: patients, laboratory, surveillance, prevention, controls, quality, training, committees and documents. Each hospital uses only the modules it has been given and each user sees only what their role allows.

Who it is for

The guide is for three people: the Platform Owner, who creates the hospital and chooses what it sees; the Hospital Admin, who sets it up; and the hospital users, who use it every day.

How to read it

If you start from zero, read Chapters 1 to 4 in order. If you are already working, open Chapter 2 ("Where am I?") and go straight to your next step.

1. The journey: from day one to everyday use

1 Preparation

Who: Project lead / Platform Owner

Where: Outside the application

1. Gather the hospital's details: name, code, city and region, bed count, contact email.
2. Decide who will be the Hospital Admin and note their full name and an email they actually read. The invitation is sent there.
3. Choose a package (Chapter 4). If unsure, start with the smaller one: you can unlock modules later and nothing is lost.
4. Prepare the list of departments and the list of users with each one's role.

It is done when: You have a package, the admin's email, a department list and a user list.

2 Create the organization

Who: Platform Owner

Where: Platform Center > Organizations > New organization

1. Open the Platform Center and go to Organizations.
2. Select "New organization" and fill in identity and location.
3. Fill in the initial Hospital Admin (name and email). An invitation is sent to them.
4. Save. The organization appears in the registry.

It is done when: The organization is in the registry and the admin shows as "Pending".

3 Choose the package and modules

Who: Platform Owner

Where: Organizations > organization record > Operating profile

1. In the organization record find the "Operating profile" section.
2. Step 1: choose the package. Each package's list shows with a tick what it includes and with a lock what it does not.
3. In the chosen package, click a module's lock to unlock it (orange, bold) or its tick to lock it. The same applies to add-ons (Occupational health, Pharmacy, Prevalence survey, LIRA & AI).
4. If an orange warning appears ("Also needs: ..."), unlock the missing module too. It is a warning; it does not block saving.
5. Step 2: read the "Summary of changes" and select "Save". If you are switching modules off, confirmation is requested.

It is done when: After saving, the Summary reads "No change from the saved profile".

4 Activate the admin's account

Who: Hospital Admin

Where: Invitation email > activation page

1. Open the invitation email and follow the link.
2. Set a personal password. Only you set it: neither the Platform Owner nor another admin can see it.
3. Select "Go to sign in" and sign in with the username shown.
4. If no email arrived: the Platform Owner opens the organization record, finds the Hospital Admin and selects "Resend invitation".

It is done when: The admin signs in and the organization record shows "Active".

5 Set up the hospital

Who: Hospital Admin **Where:** Management Center

1. Libraries > Departments: create all departments first. Users are linked to a department.
2. Users & Roles: create the users, give each a role and department. Each user receives an invitation and sets their own password.
3. Libraries: review microorganisms, antibiotics and other shared values. Core items are not deleted; they are hidden or overridden.
4. Patient-days: enter them (if Indicators or Surveillance exist). They are the denominator of the indicators.
5. Structural indicators: fill in the hospital's structural items where requested.

It is done when: Departments exist and the key users have signed in at least once.

6 Go live

Who: All users **Where:** Per package — Chapter 4

1. Follow the steps of your package, in order, from Chapter 4.
2. First week: use real data only once roles and departments have been checked.

It is done when: The first real record has been entered and shows in Analysis.

7 Daily use

Who: All users **Where:** Dashboard and your role's modules

1. Start from the Dashboard: it shows only what needs attention and what you are entitled to know.
2. Open the task that is waiting and complete it. Do not re-enter data that already exists: Laboratory, Surveillance and Indicators share the same data.
3. At the end of the week check the department's pending and overdue items.

It is done when: There are no overdue tasks without a reason.

8 Analysis and reports

Who: Infection Control team, management

Where: Analysis

1. Open Analysis and choose year and period (year, half-year, quarter or month).
2. Choose the module's tab. Every unlocked module has its own tab; locked ones do not appear.
3. Optionally choose a comparison year or a department.
4. Use print or PDF export to save or send the tab's report.

It is done when: You have a PDF or printout of the report for the period.

9 Change the package or modules later

Who: Platform Owner

Where: Organizations > record > Operating profile

1. Reopen the hospital's Operating profile and unlock or lock whatever is needed.
2. Whatever is locked is hidden from every user: menu, actions, Analysis tabs. Nothing is deleted; if unlocked again it comes back as it was.
3. Changes apply to the whole hospital. If someone does not see them at once, ask them to refresh the page.

It is done when: Users see the new modules, or no longer see the locked ones.

2. Where am I? What do I do now?

If...	Then...
The hospital does not exist on the platform yet.	Stage 2: create the organization (Platform Owner).
The organization exists but no package or modules have been chosen.	Stage 3: set the Operating profile (Platform Owner).
The Hospital Admin has never signed in.	Stage 4: account activation; if no email arrived, resend the invitation.
The admin is in, but there are no departments or users.	Stage 5: set up the hospital (departments → users → libraries).
Everything is ready and real use starts.	Stage 6: follow the steps of your package (Chapter 4).
I cannot see a module or action I expected.	Chapter 8, "I cannot see a module".
I want to add or remove a module later.	Stage 9: change the package or modules (Platform Owner).
I need a report for management or EOΔY.	Stage 8: Analysis and print/PDF.

3. What each package contains

Packages are ready-made combinations of modules and work as a shortcut: after choosing a package you can unlock or lock any module separately (stage 3). Patients is always on.

Basic records. Patient records only. Laboratory and EOΔY & EARS-Net reports can be unlocked where needed.

Laboratory & infection surveillance. Laboratory records and EOΔY & EARS-Net reports, plus surveillance (HAI, isolation, therapy, outcome) and indicators.

Full infection control programme. Surveillance plus Prevention, Controls, Quality, Training, Committees and Documents.

Module	Basic records	Laboratory & infection surveillance	Full infection control programme
Patients	✓	✓	✓
Laboratory & microbiology/AMR	—	✓	✓
EOΔY & EARS-Net reports	—	✓	✓
Infection surveillance (HAI, isolation, therapy, outcome)	—	✓	✓
Indicators	—	✓	✓
Prevention & hand hygiene	—	—	✓
Controls	—	—	✓
Quality (incidents, CAPA)	—	—	✓
Training	—	—	✓
Committees & Documents	—	—	✓
Add-ons			
Occupational health	Add-on: switched on independently of the package		
Pharmacy (antimicrobial consumption)	Add-on: switched on independently of the package		
Point prevalence survey (PPS)	Add-on: switched on independently of the package		
LIRA & AI	Add-on: switched on independently of the package		

✓ Included — Locked (unlocked separately)

4. Steps per package

Package: Basic records

Who it is for: A hospital or unit that starts with the patient registry only, or wants to add modules gradually.

Goal: A correct registry of patients without duplicates, on which the other modules are added later.

STEPS IN ORDER

- 1 Choose the package** Platform Owner
In the Operating profile choose "Basic records" and save.
- 2 Departments** Hospital Admin
Management Center > Libraries > Departments: create all departments.
- 3 Users** Hospital Admin
Users & Roles: create the Infection Control Lead and those who will register patients. Give the right role and department.
- 4 First patient** Authorized user
Patients: search for the patient first. Only if not found, create a new record with the minimum required details.
- 5 Expand** Platform Owner
When you need laboratory or EOΔY/EARS-Net reports, unlock the modules in the Operating profile (see "Laboratory records" below).

FIRST-WEEK CHECK

- ☐ All users have activated their account and see the right departments.
- ☐ Searching before every new record is a habit: there are no duplicates.
- ☐ You have decided which module is added next and when.

How you know it worked: Every patient exists once and can be found at once.

Package: Laboratory & infection surveillance

Who it is for: A hospital that wants laboratory records, EOΔY/EARS-Net reports and infection surveillance with indicators, without the full control programme.

Goal: Every positive culture ends up in a surveillance episode with HAI classification, therapy, isolation and outcome, and indicators are produced automatically.

STEPS IN ORDER

- 1 Choose the package** Platform Owner
Choose "Laboratory & infection surveillance" and save.
- 2 Departments and users** Hospital Admin
Create the departments and then the users: Laboratory (samples and results), Infection Control Lead and Members (surveillance).
- 3 Patient-days** Hospital Admin
Management Center > Patient-days: enter them per department and period. Without them, indicators have no denominator.
- 4 Sample and result** Laboratory
Laboratory: open the pending queue, choose a sample, enter and validate the result and AST. The sample is linked to the patient with no second entry.
- 5 Critical results** Laboratory
For a critical result, document the communication: who was informed and when.
- 6 Surveillance episode** Infection Control team
Surveillance: open the patient's active episode. Fill in the clinical assessment and HAI classification. Microbiology comes from the Laboratory.
- 7 Therapy, isolation, outcome** Infection Control team
In the same episode record therapy, isolation measures and reassessments. Close it with an outcome only when complete.
- 8 Indicators and Analysis** Infection Control team
Indicators: check numerator and denominator. Analysis: see Surveillance & HAI, Microbiology, AMR and National surveillance.

FIRST-WEEK CHECK

- ☐ Patient-days exist for the current period.
- ☐ At least one sample has gone from the Laboratory to Surveillance without double entry.
- ☐ Active episodes have an isolation review that is not overdue.

How you know it worked: The Lead sees the period's HAI and resistance in Analysis without manual processing.

Package: Full infection control programme

Who it is for: A hospital that wants the whole infection control programme: surveillance plus prevention, controls, quality, training, committees and documents.

Goal: A complete cycle works: recording, checking, corrective action, training and documentation.

STEPS IN ORDER

- 1 Choose the package** Platform Owner
Choose "Full infection control programme" and save.
- 2 Core operation** Everyone
First do the steps of the "Laboratory & surveillance" package (previous section). The full programme includes them.
- 3 Prevention** Infection Control team
Prevention: record WHO hand hygiene observations, bundle assessments (CLABSI, CAUTI, VAP/VAE, SSI), antiseptics and waste.
- 4 Controls** Lead and Department Managers
Controls: define the controls and assign them to departments. Users perform them with a result and documentation.
- 5 Quality** Quality Manager
Quality Center: record incidents, audits and findings. For each finding open a CAPA with an owner and due date.
- 6 Training** Training managers
Training: create a programme, add participants and record completions and scores.
- 7 Committees and Documents** Committee Secretariat
Committees: schedule meetings, write minutes and decisions. Documents: take each controlled document from draft to approval and publication.
- 8 Review** Infection Control team
In Analysis review Prevention, Hand hygiene, Controls, Quality, Training and Governance and decide corrective actions.

FIRST-WEEK CHECK

- ☐ The first cycle of controls has been defined and assigned to departments.
- ☐ The first hand hygiene observation and first bundle assessment have been recorded.
- ☐ A committee meeting is scheduled and at least one controlled document is in progress.

How you know it worked: Every finding has a CAPA with an owner, every control has a result and nothing is overdue without a reason.

Common combination: Laboratory records (Basic + Laboratory + EOΔY/EARS-Net)

When: When the hospital wants the laboratory, microbiology/AMR and EOΔY & EARS-Net reports, but not infection surveillance.

1. Platform Owner: choose "Basic records" and unlock "Laboratory & microbiology/AMR" and "EOΔY & EARS-Net reports". The card shows "Custom". Save.
2. Admin: create the departments and a user with the Laboratory role.
3. Laboratory: pending queue → sample → result and AST → critical communication where needed.
4. Analysis: Microbiology, AMR / MDR-XDR, National surveillance and Reporting & data quality (data quality, EOΔY notifications, EARS-Net export).

5. The modules one by one

Patients

Always on

WHAT IT IS

The patient registry and the entry point to the surveillance record. Always on.

WHO USES IT

Infection Control team and authorized clinical roles.

ALSO NEEDS

—

ANALYSIS AND REPORT

No tab of its own: patient details never appear in Analysis.

FIRST STEP

Search for the patient first; create only if not found.

Laboratory & microbiology/AMR

Included from: Laboratory & infection surveillance

WHAT IT IS

Samples, cultures, susceptibility (AST) and communication of critical results. It is the source of truth for microbiology.

WHO USES IT

Laboratory, Infection Control team.

ALSO NEEDS

—

ANALYSIS AND REPORT

Tabs "Microbiology" and "AMR / MDR-XDR": positive cultures, MDR/XDR/PDR, critical results, departments with findings, susceptibility per organism.

FIRST STEP

Open the pending sample queue and enter the first result.

EOΔY & EARS-Net reports

Included from: Laboratory & infection surveillance

WHAT IT IS

Reports to EOΔY and EARS-Net, with a data quality check before submission.

WHO USES IT

Infection Control Lead and Members.

ALSO NEEDS

Laboratory & microbiology/AMR

ANALYSIS AND REPORT

Tabs "National surveillance" and "Reporting & data quality": EOΔY notifications, EARS-Net export and data gaps.

FIRST STEP

Open "Reporting & data quality" and fix the gaps first.

Infection surveillance (HAI, isolation, therapy, outcome)

Included from: Laboratory & infection surveillance

WHAT IT IS

The infection surveillance record: clinical assessment, HAI classification, microbiology, therapy, isolation, devices, reassessments and outcome, in one timeline.

WHO USES IT

Infection Control team, Doctor Reviewer, Link Nurse.

ALSO NEEDS

Laboratory & microbiology/AMR

ANALYSIS AND REPORT

Tabs "Surveillance & HAI" and "Antimicrobials": episodes, confirmed and probable HAI, active isolations, invasive devices, outcome, antimicrobial use.

FIRST STEP

Open a patient's active episode and fill in the clinical assessment.

Indicators

Included from: Laboratory & infection surveillance

WHAT IT IS

Indicators with a definition, numerator, denominator and period, computed from surveillance data and patient-days.

WHO USES IT

Infection Control team, management.

ALSO NEEDS

Infection surveillance (HAI, isolation, therapy, outcome)

ANALYSIS AND REPORT

No tab of its own: indicators appear on the "Indicators" screen and feed the Overview.

FIRST STEP

Enter patient-days first, then open Indicators.

Prevention & hand hygiene

Included from: Full infection control programme

WHAT IT IS

Hand hygiene (WHO), prevention bundles, antiseptics and waste.

WHO USES IT

Infection Control team and users with the matching responsibility.

ALSO NEEDS

—

ANALYSIS AND REPORT

Tabs "Prevention" and "Hand hygiene": bundle compliance, hand hygiene compliance (WHO target $\geq 80\%$), waste per department.

FIRST STEP

Enter the first hand hygiene observation.

Controls

Included from: Full infection control programme

WHAT IT IS

Controls assigned to departments and performed with a result and documentation.

WHO USES IT

Lead, Department Managers and Department Users.

ALSO NEEDS

—

ANALYSIS AND REPORT

Tab "Controls": executions, share with findings, overdue controls, executions and findings per department.

FIRST STEP

Define a control and assign it to a department.

Quality (incidents, CAPA)

Included from: Full infection control programme

WHAT IT IS

Incidents, audits, findings and CAPA (corrective and preventive actions) in a controlled flow.

WHO USES IT

Quality Manager, Infection Control team, any employee to report an incident.

ALSO NEEDS

—

ANALYSIS AND REPORT

Tab "Quality": incidents, incidents with harm, open and overdue CAPA, severity and status.

FIRST STEP

Report an incident and open a CAPA with an owner and due date.

Training

Included from: Full infection control programme

WHAT IT IS

Training programmes, participants, completions, assessments and certificates.

WHO USES IT

Training managers; an employee sees only their own training.

ALSO NEEDS

—

ANALYSIS AND REPORT

Tab "Training": completion rate, overdue assignments, average score, per department.

FIRST STEP

Create a programme and assign it to a department.

Committees & Documents

Included from: Full infection control programme

WHAT IT IS

Committees (meetings, agenda, minutes, decisions) and controlled Documents with a lifecycle from draft to publication and archiving.

WHO USES IT

Committee Secretariat, Lead, authorized users.

ALSO NEEDS

—

ANALYSIS AND REPORT

Tab "Governance": controlled documents and documents due for review, committee meetings and pending minutes.

FIRST STEP

Schedule the first meeting or create the first document.

Occupational health

Add-on

WHAT IT IS

Occupational physician visits, vaccinations and employee follow-ups. Medical data stays separate from HR.

WHO USES IT

Occupational Physician (HR does not see medical data).

ALSO NEEDS

—

ANALYSIS AND REPORT

Tab "Employees": vaccination coverage, visits, overdue follow-ups. Aggregated only, and only with the right permission.

FIRST STEP

Enter the first visit or vaccination.

Pharmacy (antimicrobial consumption)

Add-on

WHAT IT IS

Antimicrobial consumption and dispensing, restricted-use approvals, link with surveillance therapy.

WHO USES IT

Pharmacy, Infection Control team.

ALSO NEEDS

—

ANALYSIS AND REPORT

Tab "Antimicrobials": therapies, pending approval, most used agents. Shown if Surveillance or Pharmacy is on.

FIRST STEP

Review the pending antibiotic approvals.

Point prevalence survey (PPS)

Add-on

WHAT IT IS

Periodic point prevalence surveys (PPS): how many patients have HAI and how many receive antibiotics on a given day.

WHO USES IT

Infection Control team.

ALSO NEEDS

—

ANALYSIS AND REPORT

Tab "Prevalence survey": HAI prevalence, antibiotic use and comparison with the previous survey, trend per survey.

FIRST STEP

Management Center > Prevalence survey (PPS): enter the first survey.

LIRA & AI

Add-on

WHAT IT IS

The LIRA assistant and outbreak investigations. Without an AI provider, LIRA works with its built-in rules. It works only on data the user is already entitled to see.

WHO USES IT

Infection Control team, Hospital Admin (provider connection).

ALSO NEEDS

—

ANALYSIS AND REPORT

Tab "LIRA & AI": investigations (active, closed, average time to close), active clusters, distribution per organism and month.

FIRST STEP

Management Center > LIRA & AI: connect a provider (optional). Investigations are run from Management Center > Outbreak investigations.

6. Roles

Role	What it does
Platform Owner	Creates hospitals, sets package and modules, manages demos, central libraries and platform settings. Does not see patient clinical data.
Hospital Admin	Sets up the hospital: departments, users and roles, libraries, patient-days, indicators and Management Center settings.
Infection Control Lead	Leads the infection control programme: surveillance, prevention, controls, analysis.
Infection Control Member	Performs daily surveillance and prevention under the Lead.
Department Manager	Sees their department's picture and pending items and completes what is assigned.
Department User	Performs the tasks assigned to their department, with the minimum necessary information.
Link Nurse	The department's link to the infection control team.
Laboratory	Enters samples, results and AST and communicates critical results.
Doctor Reviewer	Clinically reviews surveillance episodes.
Quality Manager	Manages incidents, audits, findings and CAPA.
Committee Secretariat	Organizes committee meetings, minutes and decisions.
Pharmacy	Tracks antimicrobial consumption and approvals.
Occupational Physician	Manages employee visits and vaccinations. Has access to the medical data.
HR Office	Administrative employee registry, without access to medical data.
Staff User	Sees their own record, their own training and certificates and whatever is assigned to them.

The exact permissions of each role are shown in Management Center > Users & Roles. A role cannot use a module that is locked for the hospital.

7. Terminology

Platform terms

Term	Meaning
Organization	A hospital or clinic on the platform, with its own users and data.
Platform Owner	The platform owner. Creates organizations and sets what each one sees, without access to clinical data.
Hospital Admin	The person who sets up and manages one organization.
Operating profile	The Platform Owner's choice of what a hospital uses: package, modules and add-ons.
Package	A ready-made combination of modules: Basic records, Laboratory & surveillance, Full programme. It is a shortcut, not a limit.
Module	A part of the platform, e.g. Laboratory or Controls, switched on or off independently.
Add-on	A module outside the packages: Occupational health, Pharmacy, Prevalence survey, LIRA & AI.
Lock / Unlock	A locked module is hidden from every user of the hospital. Its data is kept.
Custom	A combination of modules that matches no package.
Prerequisite	A module that needs another to make sense, e.g. Indicators need Surveillance. The platform warns; it does not block.
Demo	Time-limited access with synthetic data only, without real patients.
Scope	The set of data a user is entitled to see: the organization and its departments.
Capability	One specific permitted action, e.g. "create surveillance". Roles are sets of capabilities.
Surveillance episode	The single record of a possible infection, from assessment to outcome.
Bundle	A group of prevention practices assessed Yes/No/N/A, e.g. CLABSI, CAUTI.
CAPA	Corrective and preventive action, with an owner and due date, opened from a finding or incident.
Patient-days	The number of days of hospital stay. The denominator of most indicators.
Critical result	A laboratory result that requires immediate notification. The notification is recorded with time and person.

Term	Meaning
ΕΟΔΥ / EARS-Net	The Greek national public health organization and the European antimicrobial resistance surveillance network to which reports are made.
Cluster / outbreak	A group of related cases that requires investigation.
Audit trail	The immutable record of significant actions: who, what, when.

Clinical terms

Term	Meaning
HAI	Healthcare-associated infection.
IPC	Infection prevention and control.
AMR	Antimicrobial resistance.
AST	Antimicrobial susceptibility testing.
MDR	Multidrug-resistant organism according to the active versioned definition.
XDR	Extensively drug-resistant according to the active versioned definition.
PDR	Pandrug-resistant organism according to the active versioned definition.
DDD	Defined daily dose; used in antimicrobial consumption indicators.
PPS	Point prevalence survey.
BSI	Bloodstream infection, according to the applicable definition.
CLABSI	Central line-associated bloodstream infection.
CAUTI	Catheter-associated urinary tract infection.
VAP	Ventilator-associated pneumonia.
Patient-days	Patient-days: sum of daily admitted-patient counts for a defined period/department.
Device-days	Days of exposure to a specific device, used as denominator for device-associated indicators.

8. Common problems

I cannot see a module or action.

- First check whether the module is unlocked for the hospital (ask the Platform Owner): if locked, nobody sees it.
- If unlocked, check your role and permissions in Management Center > Users & Roles.
- Refresh the page if the change was made just now.

The user did not receive the invitation.

- Check the spam folder and that the email address is correct.
- Management Center > Users & Roles: open the user ("User access" dialog) and choose to resend the invitation. For the Hospital Admin themselves, the Platform Owner does it from the organization record.

I forgot my password.

- On the sign-in page use the account recovery link: instructions are sent only to the email registered to your account.
- Alternatively, the Hospital Admin resets your password from Users & Roles > "User access". For the admin, the Platform Owner does it.

If I lock a module, is data lost?

- No. Data is hidden, not deleted. If you unlock the module again, it comes back as it was.

The platform warns me: "Also needs: ...".

- The module you unlocked needs another one that is still locked. Unlock that too. You can save without it, but the module will have little or no data.

An Analysis tab is empty.

- Check the period and department you selected.
- Check that data has been entered in the matching module for that period. Indicators also need patient-days.

I want to try the platform without real patients.

- Ask the Platform Owner for a Demo: it has synthetic data only and expires automatically. The preview in the Help Center also shows the real screens with demo data.